Commun Nonlinear Sci Numer Simulat 163 (2026) 110517

Contents lists available at ScienceDirect

Communications in Nonlinear Science and Numerical Simulation

journal homepage: www.elsevier.com/locate/cnsns

Delayed full feedback control with difference between two past states and periodic gain for stabilization and chaos control

Verónica Estela Pastor [a,*], Graciela Adriana González [a,b]

[a] Universidad de Buenos Aires, Facultad de Ingeniería, Departamento de Matemáticas, Argentina
[b] Consejo Nacional de Investigaciones Científicas y Técnicas, CONICET, Argentina

ARTICLE INFO

Keywords:
Delayed feedback control
Periodic gain
Equilibrium point stabilization
Odd number limitation
Chaos control

ABSTRACT

A delayed feedback control based on full state is proposed. The designed scheme combines the difference between two delayed states and a periodic control gain. Choosing adequate control parameters values, system stabilization is achieved in any hyperbolic unstable equilibrium point so in particular, odd number limitation is overcome. The procedure to build the control is systematic, and the set of the stabilizing control parameters is rigorously featured by analytical arguments. An ad-hoc strategy based on the proposed scheme is implemented for control of chaos by stabilizing equilibrium points embedded in chaotic attractors. Simulated examples illustrate this strategy performance.

1. Introduction

Since Pyragas proposed the delayed feedback control (DFC) method in 1992 ([1]), there has been increasing interest in stabilizing unstable periodic orbits (UPO) via this resource. A distinct feature of DFC and its modifications is that it does not require preliminary knowledge about the desired UPO except its period. The UPO of the free system remains as a periodic orbit of a controlled system by a DFC method. These algorithms are noninvasive that is, the control perturbation vanishes whenever the system settles on the target orbit. The extended DFC (EDFC) was introduced for dealing with highly unstable orbits in [2] followed by many works based on it. More recently, a dynamic EDFC has been presented in [3] for stabilizing multi-rotation UPOs by enlarging the stability regions. The theoretical analysis of the DFC method is difficult because the resulting dynamics evolves in an infinite dimensional phase space. The introduction of a periodic control gain, based on the act-and-wait concept, makes the stability study hing on a finite dimensional monodromy matrix. A great deal of works on this line are referred in [4] and a more recent paper ([5]) advances on numerical aspects for obtaining stability conditions of the closed loop system. As a further development of the EDFC, it is worth to mention [6] where a state-dependent switch is appended to the feedback loop.

The DFC method has been reformulated to stabilize a given dynamical system in one of its unstable equilibrium points. It is designed as an amplified difference between the present state $x(t)$ and the delayed state $x(t-\tau)$, or between a suitable observable of the state at times t an $t-\tau$, so depending on the time delay τ and a control gain K. The EDFC version includes not only τ but also its integer multiples and its design is based on a geometric sum of feedback signals ([7]). Later, the multiple DFC (MDFC) which depends on n different delay $\tau_1, \tau_2, \ldots, \tau_n$ and different gains $k_1, k_2, \ldots, k_n$ was proposed in [8] as an alternative that provides greater stability and flexibility over its antecedents.

* Corresponding author.
E-mail address: vpastor@fi.uba.ar (V.E. Pastor).

https://doi.org/10.1016/j.cnsns.2026.110517

Available online 7 July 2026
1007-5704/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

There is a great deal of work about applications of DFC or its variants on technical problems especially for removing unwanted dynamics, for example, in delay-coupled neurosystems and in lasers ([9]) or in glycolytic reaction models ([10]) as many others cited in [3,11] or [12]. Some of the most recent are [12–14]. EDFC is tested in [12] both numerically and experimentally on an impact oscillator system; this work does not end with the application but also analyzes the loose of this method efficacy to stabilize a target UPO when delays are even multiples of the UPO's period. For suppressing undesired chaotic oscillations in an indirect field oriented control of a 3-phase induction motor, a simple DFC controller is confronted with a sliding mode DFC controller in [13]. In economy, eliminating unstable and irregular behaviour is of interest just because such scenarios are unfavourable for state prediction, as the case from [14] of the dynamics of a mid-size firm modeled by a set of differential equations for which chaos control is achieved by DFC method.

A system affected by DFC, EDFC or MDFC involves a nonlinear delayed-differential equation, so its stability analysis turn out rather intricate. Adequate values of the control parameters may be obtained by numerical computation but an analytical determination of them results a critical point. In some works ([15,16]), the stabilizing parameters are obtained by solving some linear matrix inequalities. The proposal of [15] deals with a delayed full state feedback control so the control gain K is an $n \times n$ matrix, being n the phase space dimension. An analytical procedure based on inequalities is presented in [16] to obtain the set of stability parameters of a particular case of the MDFC.

On the other hand, if the system linearization in the equilibrium has either a zero eigenvalue or an odd number of real positive eigenvalues, there are no control parameters values such that stabilization is achieved by any of the referred methods. This restriction is known as the odd number limitation (ONL) (see [17–19] for details).

The introduction of a periodic control gain allows to study the dynamics of the DFC by means of a discrete-time system. A scheme including a periodic control gain appears in [20], based on the act-and-wait-concept. It is fully developed for the unstable focus case but the ONL is not dealt with by this proposal. Later, a DFC algorithm based on a periodic control gain has been developed in [21] that overcomes the ONL although it only considers the case of saddles which eigenvalues have unit geometric multiplicity.

In [11], the classical difference of the DFC is replaced by the difference between the two delayed states $x(t-2\tau)$ and $x(t-\tau)$. For the general nonlinear scalar case, it has been proved in [11] that stabilization is achieved by this scheme combined with a periodic control gain. The periodic control law is quite simple, and the adequate control parameters can be computed systematically.

The present work extends the proposal of [11] to the n-dimensional case by incorporating the full state into the feedback. By this proposal, system stabilization is achieved in any hyperbolic unstable equilibrium point (saddle, focus or saddle-focus) so in particular, ONL is overcome. Moreover, the procedure for building the control is systematic, and the set of the stabilizing control parameters is rigorously featured by analytical arguments. Finally, the issue of controlling chaos is considered. An ad-hoc strategy based on the proposed scheme is introduced for stabilizing equilibrium points embedded in chaotic attractors.

The paper is organized as follows: Section 2 states the general problem and introduces the control design. The nonlinear local stability is based on an ad-hoc built map introduced in Section 3 and the stability control parameters are obtained. This computation is performed for the two-dimensional case in Section 3.1 and generalized to the n-dimensional case in Section 3.2. The main theorem summarizing the complete stabilization result is presented in Section 4. With these tools at hand, Section 5 implements a control chaos strategy with its performance illustrated in well-known chaotic systems. In Sections 6 and 7 the concluding remarks and an outline of future research are detailed.

2. Problem statement

Let us consider a system given by:

$$\dot{x} = f(x) \tag{1}$$

where $x \in \mathbb{R}^n$ is the state vector and $f : \mathbb{R}^n \to \mathbb{R}^n$ is a piece-wise continuously differentiable function. In particular f is locally continuously differentiable on x^*, an unstable hyperbolic equilibrium point of the system (1).

The objective is to stabilize the system in x^* by applying additive control, that is,

$$\dot{x} = f(x) + u(t) \tag{2}$$

with $u(t)$ a time-delayed feedback control.

The following control law emerges as a generalization to the n-dimensional case of the proposal of [11] for the scalar case:

$$u(t) = K(t)\big(x(t-2\tau) - x(t-\tau)\big) \tag{3}$$

where $\tau > 0$ and

$$K(t) = \begin{cases} 0_{n\times n} & \text{if } 3k\tau \leqslant t < (3k+2)\tau \\ K & \text{if } (3k+2)\tau \leqslant t < (3k+3)\tau \end{cases} \qquad k \in \mathbb{N} \cup \{0\}, \tag{4}$$

being $0_{n\times n}$ the null matrix in $\mathbb{R}^{n\times n}$ and K a constant matrix in $\mathbb{R}^{n\times n}$ which will be chosen adequately. This control strategy will be referred as D_2FC.

The existence of solution of the system (2)-(3) is justified by well known analytic tools. Note that x^* remains as an equilibrium point of (2)-(3).

The problem is to find K and τ such that x^* becomes an asymptotic stable equilibrium point of the controlled system (2)-(3).

Putting $\delta x = x - x^*$, the system (2)-(3) results:

$$\delta\dot{x} = A\delta x + g(\delta x) + K(t)\big(\delta x(t-2\tau) - \delta x(t-\tau)\big) \tag{5}$$

where $A = Df(x^*)$ is the jacobian matrix of f in x^* and g is the function that comprises the nonlinear terms with $g(0) = 0$.

In turn, there exists an invertible matrix V such that

$$A = VJV^{-1}$$

being J the Jordan matrix of A. Let us point out that, as x^* is hyperbolic, each eigenvalue of A (or, equivalently, of J) has real part different from zero.

Introducing the coordinate change:

$$y = V^{-1}\delta x$$

and defining:

$$\widetilde{K} = V^{-1}KV \tag{6}$$

system (5) is re-written as:

$$\dot{y} = Jy + V^{-1}g(Vy) + \widetilde{K}(t)\big(y(t-2\tau) - y(t-\tau)\big) \tag{7}$$

where

$$\widetilde{K}(t) = \begin{cases} 0_{n\times n} & \text{if } 3k\tau \leqslant t < (3k+2)\tau \\ \widetilde{K} & \text{if } (3k+2)\tau \leqslant t < (3k+3)\tau \end{cases} \qquad k \in \mathbb{N}\cup\{0\}.$$

The origin is an equilibrium point of the controlled system (7) so that the problem reformulates as finding the values τ and $\widetilde{K}$ that stabilize it in the origin. Then, by applying equation (6), suitable τ and K are obtained.

3. Stability control parameters computation

The controlled system (7) is a nonautonomous dynamical system, periodic but discontinuous with respect to the variable t. Then, a procedure analogous to the one used in the scalar case ([11]) may be developed for obtaining the values τ and $\widetilde{K}$ that stabilize it in the origin.

The key point is the map P in $\mathbb{R}^n$ determined by

$$P(y_{3k}) = y_{3k+3}$$

where $y_{3k} = y(3k\tau) = \lim_{t\to 3k\tau^-} y(t)$, $k \in \mathbb{N}$. The origin is a fixed point of P. As f is locally continuously differentiable on x^*, P is so on the origin. Then local stability conditions of it are obtained from its linearization and it is locally exponentially stable for P if $\rho(P'(0)) < 1$ being $\rho(P'(0)) = \max\{|\lambda|, \lambda \text{is eigenvalue of } P'(0)\}$ ([22]). By the last, the local exponentially stability of P in the origin yields -as in the scalar case- to the local asymptotic stability of system (7) in the origin.

Hence suitable τ and $\widetilde{K}$ will be obtained from $\rho(P'(0)) < 1$. In turn, $P'(0)$ is obtained from the variational equation of (7), resulting not affected by the nonlinear terms of (7). Stability control parameters to determine the matrix $\widetilde{K}$ will be computed in what follows.

3.1. The two-dimensional case

Three cases will be analyzed by considering the three possible Jordan forms of the matrix A:

$$(1)\ J = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad (2)\ J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \quad (3)\ J = \begin{bmatrix} \mu & -\omega \\ \omega & \mu \end{bmatrix}.$$

Case 1. As x^* is hyperbolic, both λ_1 and λ_2 are different from zero and as it is an unstable equilibrium point, at least one of the diagonal values are greater than zero. Proposing:

$$\widetilde{K} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix},$$

it comes out:

$$P'(0) = \begin{bmatrix} e^{3\lambda_1\tau} + k_1\tau e^{\lambda_1\tau}(1 - e^{\lambda_1\tau}) & 0 \\ 0 & e^{3\lambda_2\tau} + k_2\tau e^{\lambda_2\tau}(1 - e^{\lambda_2\tau})) \end{bmatrix}.$$

Let us fix real values $\zeta_i : |\zeta_i| < 1$, $i = 1, 2$. Putting

$$k_1 = \begin{cases} 0 & \text{if } \lambda_1 < 0 \\ \epsilon_1 & \text{if } \lambda_1 > 0 \end{cases} \quad \text{and} \quad k_2 = \begin{cases} 0 & \text{if } \lambda_2 < 0 \\ \epsilon_2 & \text{if } \lambda_2 > 0 \end{cases}$$

where

$$\epsilon_i = \frac{e^{-\lambda_i \tau}(e^{3\lambda_i \tau} - \zeta_i)}{\tau(e^{\lambda_i \tau} - 1)} \quad i = 1, 2. \tag{8}$$

both eigenvalues of $P'(0)$ are of module less than one.

Case 2. From the instability of x^*, it results $\lambda > 0$. Proposing:

$$\widetilde{K} = \begin{bmatrix} \epsilon & 0 \\ 0 & \epsilon \end{bmatrix}$$

it comes out:

$$P'(0) = \begin{bmatrix} e^{3\lambda\tau} + \epsilon\tau e^{\lambda\tau}(1 - e^{\lambda\tau}) & 3\tau e^{3\tau\lambda} + \epsilon\tau^2(1 - 2e^{2\lambda\tau}) \\ 0 & e^{3\lambda\tau} + \epsilon\tau e^{\lambda\tau}(1 - e^{\lambda\tau}) \end{bmatrix}.$$

By fixing a real value $\zeta : |\zeta| < 1$ and computing ϵ as

$$\epsilon = \frac{e^{-\lambda\tau}(e^{3\lambda\tau} - \zeta)}{\tau(e^{\lambda\tau} - 1)},$$

ζ is a double eigenvalue of $P'(0)$.

Case 3. This is the case of complex eigenvalues. As x^* is unstable, $\mu > 0$. Proposing

$$\widetilde{K} = \begin{bmatrix} \epsilon^1 & -\epsilon^2 \\ \epsilon^2 & \epsilon^1 \end{bmatrix},$$

let us introduce:

$$z = y_1 + iy_2, \quad \lambda = \mu + \mathrm{i}\omega.$$

The methodology of [11] is extrapolated and the map P, defined on the complex plane given by

$$P(z(3k\tau)) = z((3k + 3)\tau)$$

is associated as in the scalar real case. It results that

$$P'(0) = e^{3\lambda\tau}[1 + \epsilon\tau e^{-2\lambda\tau}(1 - e^{\lambda\tau})]$$

with $\epsilon = \epsilon^1 + i\epsilon^2$. Putting $P'(0) = \rho e^{i\theta}$, it is deduced that

$$\begin{aligned} \epsilon^1 &= \frac{\rho\cos(\theta - \omega\tau) - \rho e^{\mu\tau}\cos(\theta - 2\omega\tau) - e^{3\mu\tau}\cos(2\omega\tau) + e^{4\mu\tau}\cos(\omega\tau)}{\tau e^{\mu\tau}(1 - 2e^{\mu\tau}\cos(\omega\tau) + e^{2\mu\tau})}, \\ \epsilon^2 &= \frac{\rho\sin(\theta - \omega\tau) - \rho e^{\mu\tau}\sin(\theta - 2\omega\tau) - e^{3\mu\tau}\sin(2\omega\tau) + e^{4\mu\tau}\sin(\omega\tau)}{\tau e^{\mu\tau}(1 - 2e^{\mu\tau}\cos(\omega\tau) + e^{2\mu\tau})} \end{aligned} \tag{9}$$

so fixing ρ, θ: $0 \leqslant \rho < 1$ and $0 \leqslant \theta < 2\pi$, adequate ϵ^1 and ϵ^2 are obtained.

3.2. The n-dimensional case

Let us go back to the general n-dimensional problem under the hypothesis and notation of Section 2. Assume that the matrix A has n_r real eigenvalues $\lambda_1, \ldots, \lambda_{n_r}$ and $2n_c$ complex eigenvalues $\lambda_{n_r+1}, \overline{\lambda_{n_r+1}}, \ldots, \lambda_n, \overline{\lambda_n}$, all of them counted with their multiplicity. Then, $n = n_r + 2n_c$ with $n_r \geqslant 0$ and $n_c \geqslant 0$ and the Jordan matrix of A may be written as

$$J = \begin{bmatrix} \lambda_1 & \delta_1 & 0 & \cdots & & & & & 0_{1\times 2} \\ 0 & \ddots & \ddots & \ddots & & & & & \\ \vdots & \ddots & \lambda_{n_r-1} & \delta_{n_r-1} & & & & & \\ \vdots & & \ddots & \lambda_{n_r} & 0_{1\times 2} & & & & \\ 0_{2\times 1} & & & & B_{n_r+1} & \delta_{n_r+1} I_{2\times 2} & & & \\ \vdots & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & & & & B_{n-1} & \delta_{n-1} I_{2\times 2} \\ 0_{2\times 1} & & & & & & & & B_n \end{bmatrix} \tag{10}$$

where $\delta_i = 0$ or 1, $0_{i\times j}$ are the null $i \times j$-vectors, $I_{2\times 2}$ is the identity 2×2-matrix and

$$B_i = \begin{bmatrix} \mu_i & -\omega_i \\ \omega_i & \mu_i \end{bmatrix}$$

with $\mu_i = \mathrm{Re}(\lambda_i)$, $\omega_i = \mathrm{Im}(\lambda_i)$ for $i = n_r + 1, \cdots, n$.

The matrix $\widetilde{K}$ is proposed as the following generalized diagonal matrix:

$$\widetilde{K} = \begin{bmatrix} k_1 & & & & & 0_{1\times 2} \\ & \ddots & & & & \\ & & k_{n_r} & & & \\ & & & E_{n_r+1} & & \\ & & & & \ddots & \\ 0_{2\times 1} & & & & & E_n \end{bmatrix} \tag{11}$$

where

$$k_i = \begin{cases} 0 & \text{if } \lambda_i < 0 \\ \epsilon_i & \text{if } \lambda_i > 0 \end{cases} \text{ and } E_i = \begin{cases} 0_{2\times 2} & \text{if } \mathrm{Re}(\lambda_i) < 0 \\ \begin{bmatrix} \epsilon_i^1 & -\epsilon_i^2 \\ \epsilon_i^2 & \epsilon_i^1 \end{bmatrix} & \text{if } \mathrm{Re}(\lambda_i) > 0 \end{cases} \quad i = n_r + 1, \dots, n.$$

The map P results from the extension and combination of the cases developed in Section 3.1. The generalized diagonal entries of $P'(0)$ preserve the dimension of the corresponding generalized diagonal entries of $\widetilde{K}$. Moreover, $P'(0)$ results a generalized diagonal matrix if $\delta_i = 0$ for all i (as in Case 1) while it is a generalized upper triangular matrix if any δ_i equals 1 (as in Case 2). Hence, in any case its eigenvalues are obtained from its diagonal entries and the formulae of the adequate control parameters is deduced, resulting ϵ_i as in (8) for $i = 1, \dots, n_r$ and ϵ_i^1 and ϵ_i^2 as ϵ^1 and ϵ^2 in (9) but replacing μ, ω, ρ and θ by μ_i, ω_i, ρ_i and θ_i respectively with $0 \leqslant \rho_i < 1$, $0 \leqslant \theta_i < 2\pi$ for $i = n_r + 1, \dots, n$.

Remark 1. Let us note that despite the δ_i values affect the dynamics (7), the computation of $\widetilde{K}$ does not hold on them. This means that the controller design ignores the non-diagonal elements in the Jordan blocks, which may lead to decreased control performance.

4. Control strategy D_2FC for stabilization

The design of the control strategy D_2FC has been developed throughout the Sections 2 and 3. Formulae for computing the stabilizing control parameters have been deduced. Its achievement has been justified and may be summarized as follows.

Theorem 1. *Let x^*, an unstable hyperbolic equilibrium point of the system (1) with $f : \mathbb{R}^n \to \mathbb{R}^n$ is a piece-wise continuously differentiable function and locally continuously differentiable on x^*.*

Assume that $n = n_r + 2n_c$ with $n_r \geqslant 0$ and $n_c \geqslant 0$ and that $Df(x^*)$, the jacobian matrix of f in x^*, has n_r real eigenvalues $\lambda_1, \dots, \lambda_{n_r}$ and $2n_c$ complex eigenvalues $\lambda_{n_r+1}, \overline{\lambda_{n_r+1}}, \dots, \lambda_n, \overline{\lambda_n}$, all of them counted with their multiplicity. Besides, consider the decomposition:

$$Df(x^*) = VJV^{-1}$$

being J the real Jordan form of $Df(x^*)$ given by (10).

If $K = V\widetilde{K}V^{-1}$ and, $\widetilde{K}$ is computed by formulae (11), then x^* is an asymptotic stable equilibrium point of:

$$\dot{x} = f(x) + K(t)\big(x(t-2\tau) - x(t-\tau)\big)$$

with $K(t)$ defined by (4).

5. D_2FC for controlling chaos

Consider that the system (1) is chaotic. The tools developed in the previous sections may be implemented for controlling chaos by introducing a strategy based on the algorithm (3)-(4). Let δ be such that for all initial condition within a δ-distance from the equilibrium x^*, asymptotic stabilization in x^* is achieved by applying (3)-(4). The strategy consists in two steps. In the first stage, the control is not activated (*i.e.*, the system runs free). The second stage begins at a certain time t^* for which the trajectory $x(t)$ satisfies $||x(t) - x^*|| < \delta$ and control is activated. Namely,

$$u_c(t) = \begin{cases} 0 & \text{if } 0 \leqslant t \leqslant t^* \\ K_c(t)\big(x(t-2\tau) - x(t-\tau)\big) & \text{if } t \geqslant t^* \end{cases} \tag{12}$$

where $t^* = inf\{t \geqslant 2\tau, ||x(t) - x^*|| \leqslant \delta\}$ and $K_c(t) = K(t - t^*)$ for $K(t)$ defined in (4).

The existence of t^* (usually called "wait time") can be argued under the chaotic attractor ergodicity as referred in [23]. However, this is valid only under additional measure-theoretic assumptions and usually only works for almost initial point of the attractor. Besides, a small neighbourhood of a given point of the attractor may be visited so infrequently that the free dynamics takes a long time (*i.e.* unrealistic) to approach the desired target.

Effective implementation of this strategy requires the target equilibrium point to be embedded in the attractor, which demands knowledge of its basin of attraction (*i.e.* a trapping region). The wait time is computed online with a stop criterion to deal with unimplementable cases.

For illustration, the strategy (12) will be used to control chaos of the classical chaotic systems known as Chua's and Rossler systems in some of its equilibrium. Simulated trajectories will be appreciated both in three dimensional graphs and by their respective state and control signals. In the electronic version of this paper, the green colour belongs to the "wait time" period and once the control is activated, red and blue are used for free system and controlled system respectively.

5.1. Chua's equation

Chua's circuit is a classical device to study chaos phenomena. The circuit resulting from adding a linear resistor in series with the inductor has been named as Chua's oscillator which presents different dynamical features for certain variations on its parameters ([24]). Its dimensionless equation is given by:

$$\begin{cases} \dot{x_1} = \alpha\big(x_2 - x_1 - f(x_1)\big) \\ \dot{x_2} = x_1 - x_2 + x_3 \\ \dot{x_3} = -\beta x_2 - \gamma x_3 \end{cases} \tag{13}$$

where

$$f(x_1) = m_1 x_1 + \frac{1}{2}(m_0 - m_1)\big(|x_1 + 1| - |x_1 - 1|\big). \tag{14}$$

Let us note that (13)-(14) may be re-written as the following piece-wise linear differential equation

$$\dot{x} = \begin{cases} A_1 x(t) - d & \text{if } x \in S_{-1} = \big\{(x_1, x_2, x_3) : x_1 \leqslant -1\big\} \\ A_2 x(t) & \text{if } x \in S_0 = \big\{(x_1, x_2, x_3) : |x_1| \leqslant 1\big\} \\ A_1 x(t) + d & \text{if } x \in S_1 = \big\{(x_1, x_2, x_3) : x_1 \geqslant 1\big\} \end{cases} \tag{15}$$

with

$$A_1 = \begin{bmatrix} -\alpha(1+m_1) & \alpha & 0 \\ 1 & -1 & 1 \\ 0 & -\beta & -\gamma \end{bmatrix},\ A_2 = \begin{bmatrix} -\alpha(1+m_0) & \alpha & 0 \\ 1 & -1 & 1 \\ 0 & -\beta & -\gamma \end{bmatrix},\ d = \begin{pmatrix} \alpha(m_0 - m_1) \\ 0 \\ 0 \end{pmatrix}.$$

In the general case, system (15) presents three equilibrium points and each one belongs to each S_i. In particular, the origin is one of them. By varying the parameters set, Chua's equation presents different analytical and geometric features ([24]).

Chua's oscillator reduces to Chua's circuit upon setting the linear resistor R_0 to zero, *i.e.*, $\gamma = 0$ which includes the cases known as double scroll and the double hook because of the aspect of their respective attractors. For both of them, the origin is in the attractor and it is an equilibrium point having only one positive eigenvalue, although the linearization in the origin of the double scroll has one real eigenvalue and two complex eigenvalues while for the double hook has three real eigenvalues.

In [25], a rigorous ad-hoc algorithms are developed to compute solutions of continuous piece-wise linear systems. These tools are used to obtain trapping regions encompassing the scroll and hook attractors which are required to implement the control strategy (12) on the aforementioned Chua's circuits. Furthermore, these tools are effective in computing trajectories that pass too close to the origin, a feature that must be taken into account if (12) is to be applied with δ very small.

5.1.1. Double scroll stabilized in the origin

The study on the double scroll attractor dates back to the middle eighties ([26]) and it was sustained by a great deal of work afterwards, but a rigorous analysis of its complex structure remains open ([27]). The case given by (15) with $\alpha = 9.3515$, $\beta = 14.79$, $\gamma = 0$, $m_0 = -1.138$ and $m_1 = -0.722$ is considered in [25] where a trapping region given by the union of four polygons is identified and trajectories starting in the border of one of these polygons are computed. In particular, it is shown how close to the origin the trajectory initiated in $CI = (1.5,\ -0.253849008275,\ -2.55651050226)$ passes.

The objective is to control chaos by stabilizing the system in the origin. The linearization matrix is given by A_2 which has the eigenvalues: $\lambda_1 \approx 2.2407$, $\lambda_{2,3} \approx -0.9751 \pm 2.7508i$. Hence the origin is a saddle-focus equilibrium.

The control strategy (12) is implemented considering the Jordan decomposition of A_2 given by

$$J = \begin{bmatrix} 2.2407 & 0 & 0 \\ 0 & -0.9751 & -2.7508 \\ 0 & 2.7508 & -0.9751 \end{bmatrix},\quad V = \begin{bmatrix} 0.8276 & -0.1546 & 0.4257 \\ 0.0841 & 0.1627 & -0.0577 \\ -0.5550 & -0.8747 & -0.8747 \end{bmatrix}.$$

Then, taking $\zeta = 0.4.$ and $\tau = 0.1$, it is obtained:

$$\bar{K} = \begin{bmatrix} 49.5974 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},\quad K = \begin{bmatrix} 46.8954 & 123.5023 & 14.6824 \\ 4.7652 & 12.5495 & 1.4919 \\ -31.4526 & -82.8327 & -9.8475 \end{bmatrix}.$$

Putting $\delta = 1.8$, the results from simulation of the controlled system with CI as the initial condition are appreciated in Figs. 1 and 2. The free and controlled system initiated in CI are depicted in Fig. 1(a) and Fig. 1(b) respectively, for confrontation. The δ-sphere centered at the origin depicted in both graphs put in evidence the effect of the control on the states bound. It can be seen in Fig. 1(b) that once the control is activated, the trajectory no longer leaves the sphere of ratio δ. The state behaviour and control performance are also appreciated by the state and control signals of the controlled system displayed in Fig. 2.

Simulation of the controlled system, keeping the ζ and τ values but choosing $\delta = 0.5$ is illustrated in Fig. 3 where the resulting trajectory in the three dimensional space and the respective temporal states and control signals are shown. Let us note that although the trajectory goes out the sphere during some time, it eventually remains there and achieves convergence.

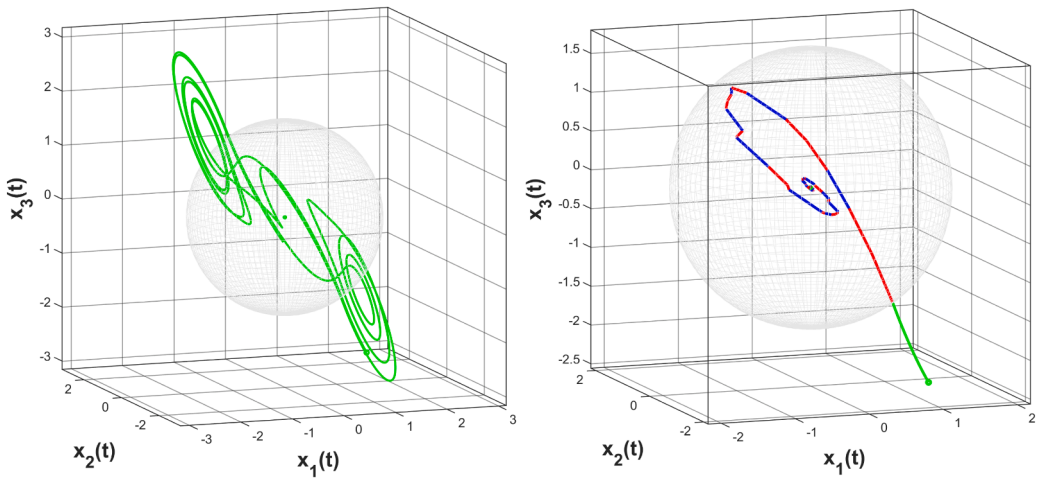


Fig. 1. Double scroll. Trajectory initialized in $CI = (1.5, -0.253849008275, -2.55651050226)$ of (a) the free and (b) the controlled system with $\zeta = 0.4$, $\tau = 0.1$ and $\delta = 1.8$.

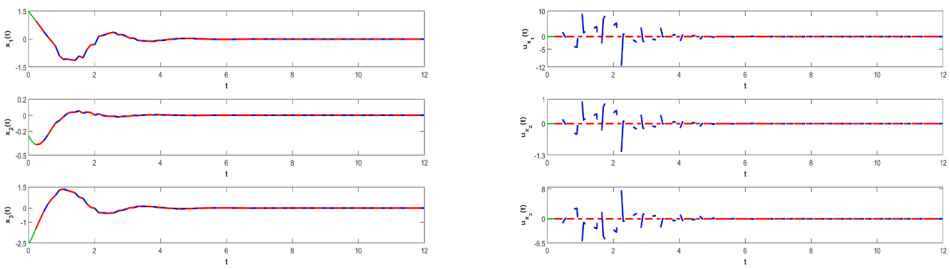


Fig. 2. Double scroll. (a) State and (b) control signals of the controlled system initialized in $CI = (1.5, -0.253849008275, -2.55651050226)$ with $\zeta = 0.4$, $\tau = 0.1$ and $\delta = 1.8$.

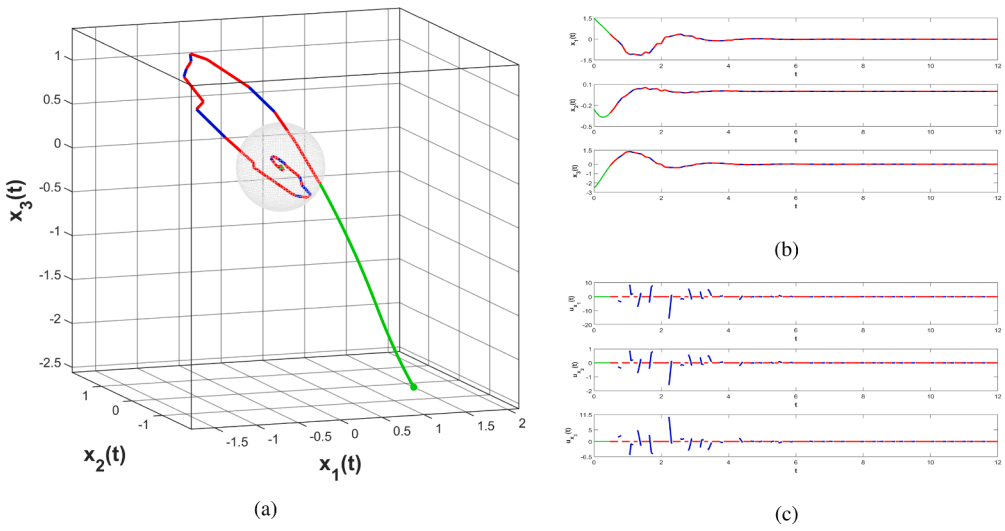


Fig. 3. Double scroll controlled with $\zeta = 0.4$, $\tau = 0.1$ and $\delta = 0.5$. Trajectory initiated in $CI = (1.5, -0.253849008275, -2.55651050226)$: (a) in 3D-space, (b) state and (c) control signals.

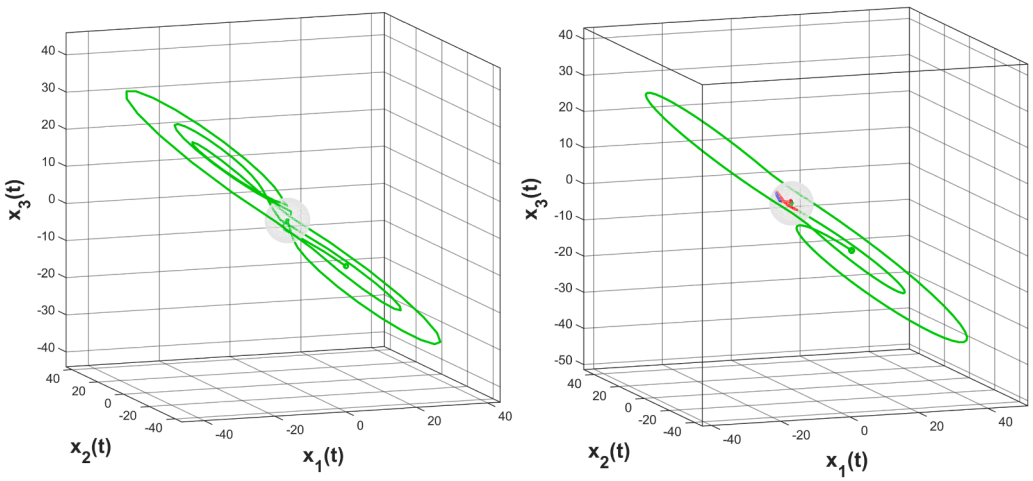


Fig. 4. Double hook. Trajectory initialized in $CI = (18,\ 1.5776491929,\ -14.3838522424)$ of (a) the free and (b) the controlled system with $\zeta = 0.4$, $\tau = 0.25$ and $\delta = 6$.

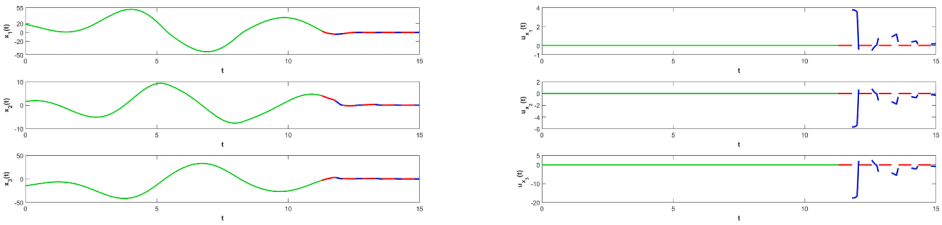


Fig. 5. Double hook. (a) State and (b) control signals of the controlled system initialized in $CI = (18,\ 1.5776491929,\ -14.3838522424)$ with $\zeta = 0.4$, $\tau = 0.25$ and $\delta = 6$.

5.1.2. *Double hook stabilized in the origin*

The double hook attractor was presented in [28] when introducing a totally different parameter set and it goes on being a subject of interest ([29]). The case considered in [25] belongs to (15) with $\alpha = -6$, $\beta = -4.442$, $\gamma = 0$, $m_0 = -2.265$ and $m_1 = -0.93$. As in the previous example, a trapping region is computed in [25] and in particular, the point $CI = (18,\ 1.5776491929,\ -14.3838522424)$ belongs to it.

The objective is to control chaos by stabilizing the system in the origin. The linearization matrix is given by A_2 which has the eigenvalues: $\lambda_1 \approx 1.4336$, $\lambda_2 \approx -3.7467$, $\lambda_3 \approx -6.2768$. Hence, the origin is a saddle equilibrium.

The control is designed taking the Jordan decomposition of A_2 given by

$$J = \begin{bmatrix} -6.2769 & 0 & 0 \\ 0 & -3.7467 & 0 \\ 0 & 0 & 1.4336 \end{bmatrix}, \quad V = \begin{bmatrix} 0.9659 & -0.7094 & -0.2001 \\ -0.2114 & 0.4544 & 0.3009 \\ 0.1496 & -0.5387 & 0.9324 \end{bmatrix}$$

and choosing $\zeta = 0.4$ and $\tau = 0.25$ which yields to:

$$\tilde{K} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 16.4102 \end{bmatrix}, \quad K = \begin{bmatrix} -0.3916 & -3.5339 & -2.4651 \\ 0.5890 & 5.3147 & 3.7073 \\ 1.8250 & 16.4678 & 11.4871 \end{bmatrix}.$$

The control strategy (12) is implemented on this system with $\delta = 6$ and it is simulated with initial condition CI. Confrontation of the free and the controlled systems are appreciated in Fig. 4, showing the respective trajectories in relation to the sphere of ratio 6. In Fig. 4(b), it can be seen that the trajectory evolves freely enough time to reveal the double hook structure. State behaviour and control performance are also appreciated through the evolution of each signal in Fig. 5. Convergence is achieved even activating the control not so close from the origin, as it happens for the same case but with $\delta = 10$ (see Fig. 6) although affecting the control performance, *i.e.* the time of convergence is reduced but in turn, it demands more control effort.

Remark 2. In the both previous examples the linearization matrix has only one positive eigenvalue so the technique of [21] also applies. Conversely, the present proposal involves less parameters and provides a full description of their admissible values.

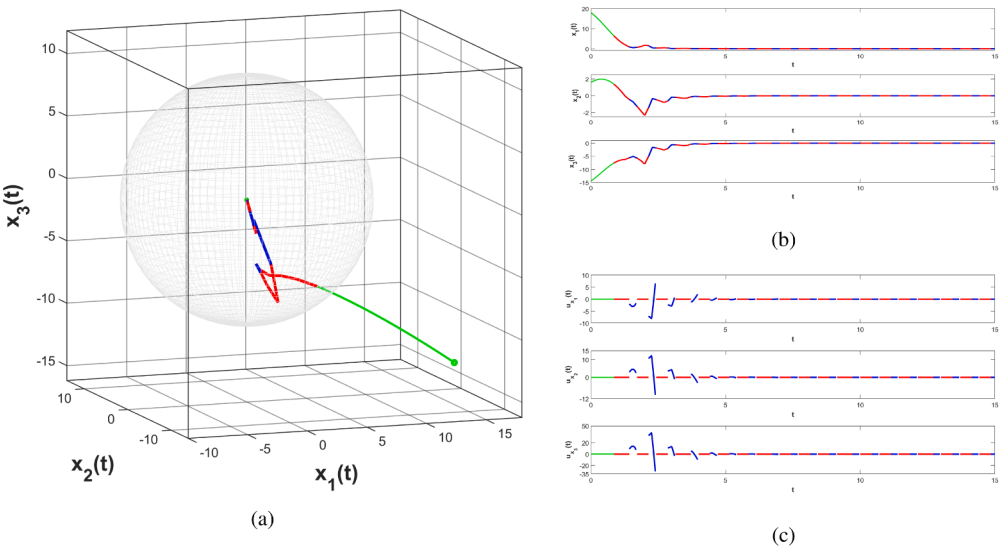


Fig. 6. Double hook controlled with $\zeta = 0.4$, $\tau = 0.25$ and $\delta = 10$. Trajectory initiated in $CI = (18,\ 1.5776491929,\ -14.3838522424)$: (a) in 3D-space, (b) state and (c) control signals.

5.1.3. *Double scroll stabilized out of the origin*

The double scroll case is revisited but for stabilizing in another of its equilibrium points. This example is taken from [20] and it coincidences with (15) for $\alpha = 9,\ \beta = 100/7,\ \gamma = 0,\ m_0 = -8/7$ and $m_1 = -5/7$. In [20], the control objective is achieved by applying a delayed feedback control based on the act-and-wait concept.

The objective is to control chaos by stabilizing the system in the equilibrium point $P = (-1.5, 0, 1.5)$ which linearization matrix has the following three eigenvalues: $\lambda_1 \approx -3.9421,\ \lambda_{2,3} \approx 0.1854 \pm 3.0470i$.

Then, by considering the Jordan decomposition given by

$$J = \begin{bmatrix} -3.9421 & 0 & 0 \\ 0 & 0.1854 & -3.0470 \\ 0 & 3.0470 & 0.1854 \end{bmatrix}, \quad V = \begin{bmatrix} -0.8678 & 0.2600 & 0.3250 \\ 0.1322 & 0.1897 & 0.0115 \\ 0.4790 & -0.8892 & -0.8892 \end{bmatrix},$$

and choosing $\tau = 0.25$, $\rho = 0.6$, $\theta = \arctan(\omega/\mu) \approx 1.5100$ being $\mu = \mathrm{Re}(\lambda_2)$; $\omega = \mathrm{Im}(\lambda_2)$, it results:

$$\tilde{K} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4.1261 & -0.6542 \\ 0 & 0.6542 & 4.1261 \end{bmatrix}, \quad K = \begin{bmatrix} -0.8437 & 0.3341 & -1.6208 \\ 1.1124 & 5.2713 & 0.5610 \\ 1.1817 & -6.1008 & 3.8247 \end{bmatrix}.$$

The control strategy (12) is applied by taking $\delta = 2$ and stabilization is achieved as it can be seen in Fig. 7. The trajectory of the controlled system with initial condition $CI = (2.104,\ -0.3188,\ -2.0866)$ in phase space is displayed in Fig. 7(a) while the state and control signals are appreciated in Figs. 7(b) and 7(c), respectively.

It is worth to point out the importance of choosing the appropriate parameter values. Indeed, the control objective is not achieved just by introducing a change in the delay value; namely fixing $\tau = 0.5$, it comes out:

$$\tilde{K} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1.7022 & -1.7121 \\ 0 & 1.7121 & 1.7022 \end{bmatrix}, \quad K = \begin{bmatrix} 0.9869 & 5.3616 & 0.3087 \\ 1.0393 & 4.0156 & 0.7751 \\ -3.6907 & -18.4419 & -1.5981 \end{bmatrix}$$

but the resulting trajectory diverges as shown in Fig. 8(a). In turn by changing the δ value, for example putting $\delta = 3$, the trajectory converges to the other equilibrium point of the system (see Fig. 8(b)).

Remark 3. This case is outside the ONL and, hence the methodology based on the act-and-wait concept also applies ([20]). The bounds of the state and control signals obtained using this method or the strategy (12) are similar (confront figure 5 of [20] and Fig. 7(b)). As the initial condition is not near enough to the equilibrium point, for the first method the state signals (see figures 4 and 6 of [20]) leave oscillatory behaviour too much later than by means of (12) (see Fig. 7(b)); this is thank to the introduction of the parameter δ (and choosing an adequate value for it).

Remark 4. Global stability and convergence are not guaranteed. This is because the stability study is based on a local stability analysis (this is also known as the linearization of the map P). The examples in Fig. 8 actually illustrate this issue.

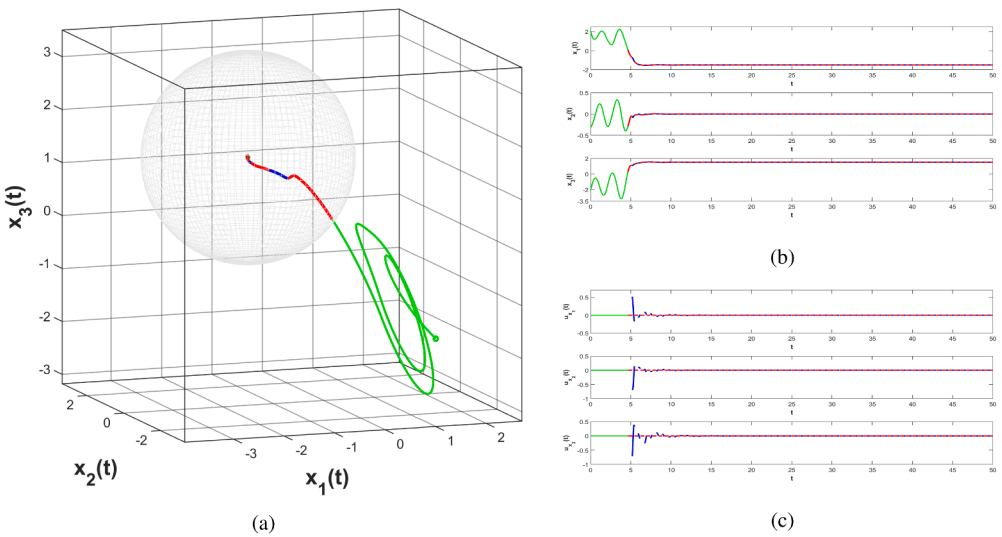


Fig. 7. Double scroll controlled with $\rho = 0.6$, $\theta \approx 1.5100$, $\tau = 0.25$ and $\delta = 2$. Trajectory initiated in $CI = (2.104,\ -0.3188,\ -2.0866)$: (a) in 3D-space, (b) state and (c) control signals.

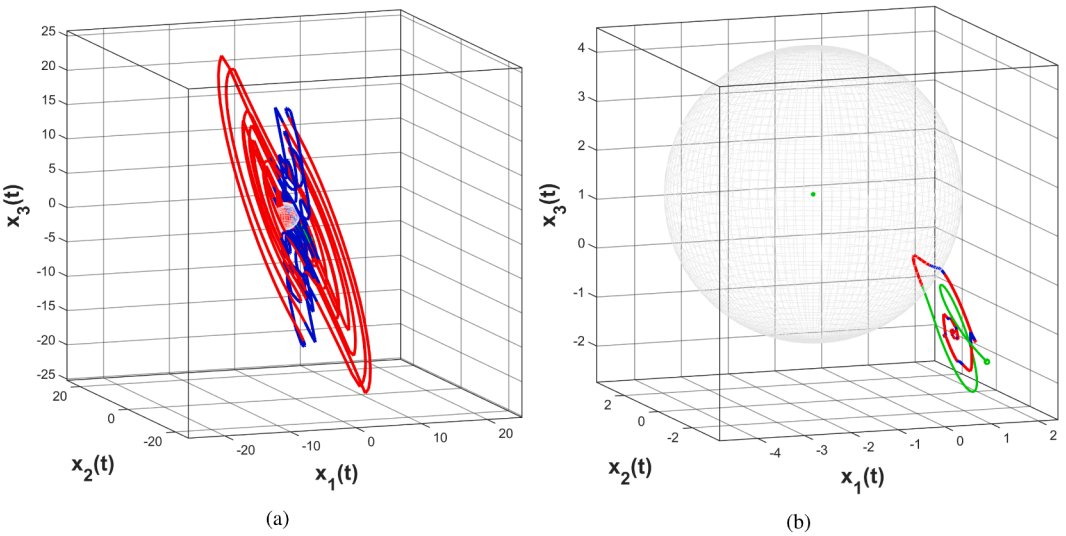


Fig. 8. Double scroll controlled in 3D-space, with $\rho = 0.6$, $\theta \approx 1.5100$, trajectory initiated in $CI = (2.104,\ -0.3188,\ -2.0866)$: (a) $\tau = 0.5$ and $\delta = 2$, (b) $\tau = 0.25$ and $\delta = 3$.

5.2. Rossler system

The well known Rossler system is a nonlinear system determined by the following set of continuously differentiable equations:

$$\begin{cases} \dot{x_1} = -x_2 - x_3 \\ \dot{x_2} = x_1 + 0.2x_2 \\ \dot{x_3} = 0.2 + x_3(x_1 - 5.7) \end{cases}.$$

For $a = 0.2$, $b = 0.2$, and $c = 5.7$, this system is chaotic. This case is considered in [15] for stabilizing it in its equilibrium point, by means of a delayed feedback control design.

Here, the objective is to control chaos by stabilizing the system in its equilibrium point $P = (0.007,\ -0.0352,\ 0.0352)$ which linearization matrix has the eigenvalues $\lambda_1 \approx -5.6940$, $\lambda_{2,3} \approx 0.0970 \pm 0.9952i$.

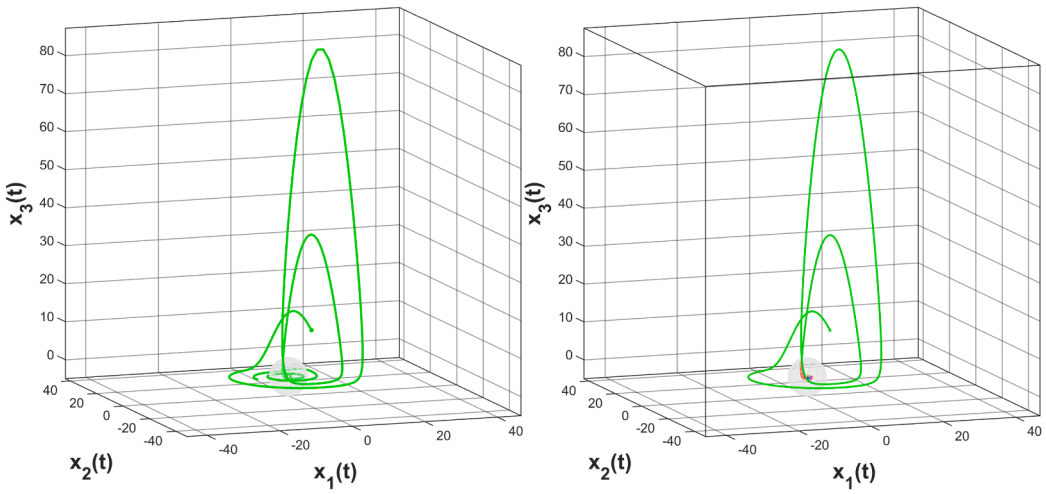


Fig. 9. Rossler. Trajectory initialized in CI $= (10, 10, 10)$ of (a) the free and (b) the controlled system with $\tau = 0.35,\ \rho = 0.2,\ \theta \approx 1.4736$ and $\delta = 5$.

For control implementation, the Jordan decomposition given by

$$J = \begin{bmatrix} -5.6940 & 0 & 0 \\ 0 & 0.0970 & -0.9952 \\ 0 & 0.9952 & 0.0970 \end{bmatrix},\ V = \begin{bmatrix} 0.1680 & 0 & 0.7073 \\ -0.0285 & -0.7032 & -0.0728 \\ 0.9854 & -0.0007 & 0.0042 \end{bmatrix}$$

is considered while chosen $\tau = 0.35$, $\rho = 0.2$, and $\theta = \arctan(\omega/\mu) \approx 1.4736$ where $\mu = \mathrm{Re}(\lambda_2)$; $\omega = \mathrm{Im}(\lambda_2)$ coming out:

$$\tilde{K} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3.6563 & 6.2539 \\ 0 & -6.2539 & 3.6563 \end{bmatrix},\ K = \begin{bmatrix} 4.3069 & 6.2911 & -0.5525 \\ -6.2911 & 3.0078 & 1.1599 \\ 0.0194 & 0.0408 & -0.0021 \end{bmatrix}.$$

Putting $\delta = 5$, the results of the simulation with initial condition $CI = (10,\ 10,\ 10)$ are displayed in Figs. 9 and 10.

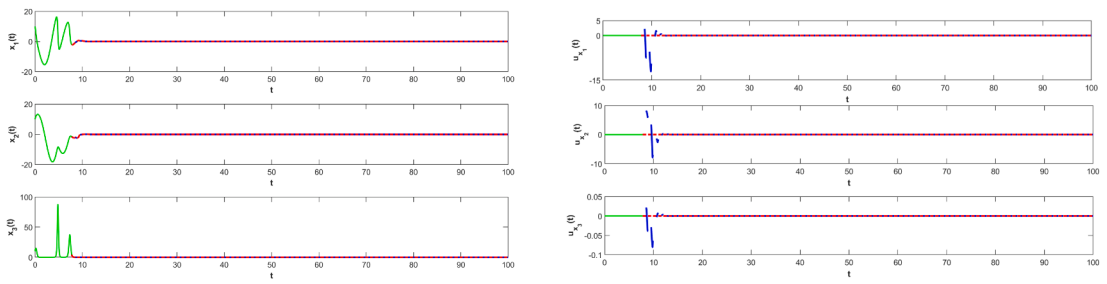
By completeness, a last example with the same parameters values but a larger time delay (*i.e.* $\tau = 5$) is given, resulting:

$$\tilde{K} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.2434 & 0.1865 \\ 0 & 0.1865 & 0.2434 \end{bmatrix},\ K = \begin{bmatrix} 0.2630 & 0.1877 & -0.0394 \\ -0.1877 & 0.2241 & 0.0385 \\ 0.0014 & 0.0014 & -0.0002 \end{bmatrix}$$

and which simulation is shown in Fig. 11.

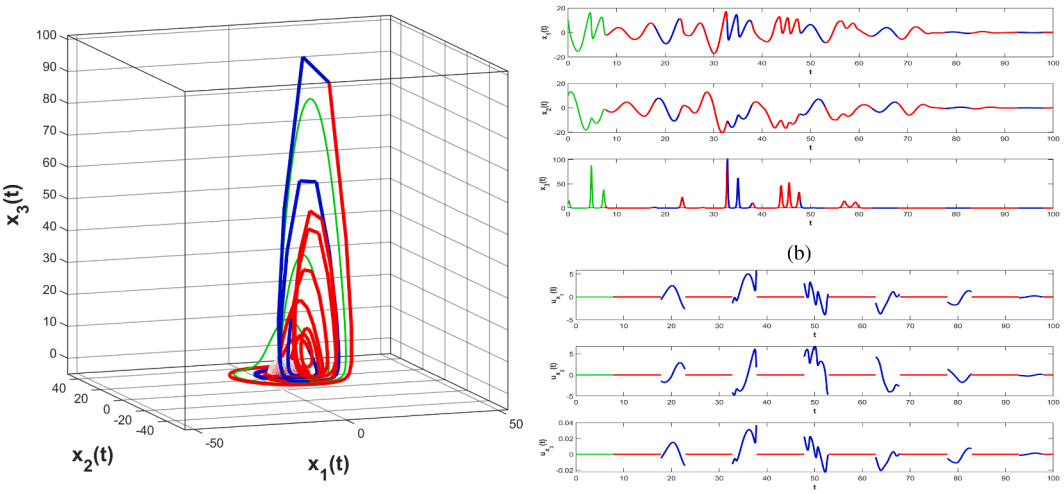
Note that when taking a larger delay time τ, even under the control regime, deteriorates the state's behaviour at the beginning, thus delaying convergence. Conversely, in the case of a small τ, the control magnitude increases significantly.

Remark 5. This example is dealt with in [15] where the control objective is satisfied by applying a full delayed feedback control. The resulting signals (see figures 7 to 10 of [15]) are comparable to those obtained by this work proposal (Figs. 9 and 10). In [15], controller gain and time delay are obtained by solving certain linear matrix inequalities. Moreover, the time delay choosing restricts to an upper bound with $\tau \leqslant 0.4183$ in [15] while for the strategy introduced here, given a desired convergence index, the control gain is computed explicitly for any time delay value (see Fig. 11, where $\tau = 5$).



(a) (b)

Fig. 10. Rossler. (a) State and (b) control signals of the controlled system initialized in CI = (10, 10, 10) with $\tau = 0.35$, $\rho = 0.2$, $\theta \approx 1.47363$ and $\delta = 5$.



(a) (b) (c)

Fig. 11. Rossler controlled system initialized in CI = (10, 10, 10) with $\tau = 5$, $\rho = 0.2$, $\theta \approx 1.47363$ and $\delta = 5$: (a) in 3D-space, (b) state and (c) control signals.

6. Concluding remarks

A control scheme for equilibrium points stabilization in the n-dimensional case has been proposed. It is a full state feedback control based on the difference between the τ and 2τ-delayed states and an adequate periodic control gain which depends on a constant matrix K.

The controlled system results a discontinuous time-delayed differential equation. Its stabilization in a desired equilibrium point is rigorously studied through an associated continuously differentiable map. It follows that given any hyperbolic equilibrium point:

- fixed any pair of stabilizing control parameters τ and K, the control objective is achieved; in particular, the ONL is overcome;
- the set of the stabilizing control parameters is analytically obtained, yielding to a systematic computation of these values;
- how the variation on the control parameters values impact on different performance aspects is looked over.

By adding a nearness-criteria through a third control parameter, δ, a strategy is implemented for controlling chaos by stabilizing equilibrium points embedded in a strange attractor. Performance features of this strategy when applied to the well known chaotic Chua and Rossler systems are observed and some improvements over previous schemes are pointed out.

7. Future work

Motivated by the aforementioned advantages of the D_2FC, novel proposals could be designed with the aim of improving control features in different scenarios:

- The issue of equilibrium point stabilization for a control system given by

$$\dot{x} = f(x,u), \quad y = g(x) \tag{16}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^m$ is the input (control) vector, $y \in \mathbb{R}^l$ is the output vector, and the functions f and g are differentiable, has been dealt with ([19,30,31]). System (16) with null control has an unstable hyperbolic equilibrium point x^*, that is $f(x^*, 0) = 0$. The objective is to stabilize the system in x^* by applying an output feedback control.

The delay feedback stabilization for two and three dimensional controllable linear systems is studied in [31] for scalar input and scalar output (*i.e.*, $m = l = 1$), based on Pyragas delayed method, that is,

$$u(t) = k\big(y(t-\tau) - y(t)\big)$$

for adequate $k \neq 0$ and $\tau > 0$. There are cases for which this method does not work, namely, the ONL remains. Hence, it sounds attractive to propose a delayed output feedback of the form:

$$u(t) = \epsilon(t)\big(y(t-2\tau) - y(t-\tau)\big)$$

with ϵ a periodic function like (4).

- In the MDFC proposal ([8,16]) two (or more delays) take part into the feedback control. It is required that the delays are not integer multiples of each other. By this method, equilibrium point stabilization is achieved for a broader region of stability parameters in comparison to DFC and EDFC. Inspired in this proposal, a suitable modification on the D_2FC could be:

$$u(t) = k(t)\big(y(t-\tau_1) - y(t-\tau_2)\big)$$

being $\tau_1 < \tau_2$ mutually prime delays and $k(t)$ an adequate periodic control gain.

- The design of a strategy as (3) for UPO´s stabilization is quite straight. Suppose that $\widetilde{x}(t)$ is a UPO which is not precisely located but its period T is known. Oscillating feedback control based on delayed states given by

$$u(t) = K(t)[x(t-2T) - x(t-T)]$$

with $K(t)$ an oscillating control gain could be proposed. As the methods developed in [4] and [32], this proposal involves a periodic control gain, but it differs from them in the difference between delayed states. In turn, an EDFC ([2]) based on the D_2FC might be built for dealing with high-instabilities scenarios. These problems, and additionally, their application for controlling chaos, by stabilizing UPO's embedded in a strange attractor are interesting for future research.

CRediT authorship contribution statement

Verónica Estela Pastor: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization; **Graciela Adriana González :** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

"I have nothing to declare".

Acknowledgements

The authors are grateful to the anonymous reviewers for their insightful comments and constructive suggestions.

Data availability

No data was used for the research described in the article.

References

[1] Pyragas K. Continuous control of chaos by self-controlling feedback. Phys Lett A 1992;170(6):421–8. https://doi.org/10.1016/0375-9601(92)90745-8
[2] Socolar JE, Sukow DW, Gauthier DJ. Stabilizing unstable periodic orbits in fast dynamical systems. Phys Rev Stat Phys Plasmas Fluids Relat Interdiscip Top 1994;50(4):23245–3248. https://doi.org/10.1103/physreve.50.3245
[3] Zheng YG, Yu JL. Stabilization of multi-rotation unstable periodic orbits through dynamic extended delayed feedback control. Chaos Solit Fract 2022;161. https://doi.org/10.1016/j.chaos.2022.112362
[4] Pyragas V, Pyragas K. Act-and-wait time-delayed feedback control of autonomous systems. Phys Lett A 2018;382(8):574–80. https://doi.org/10.1016/j.physleta.2017.12.019
[5] Liu D, Yan G. Stabilization of chaotic systems via act-and-wait delayed feedback control using a high-precision direct integration method. SN Appl Sci 2020;2,521:2523–3971. https://doi.org/10.1007/s42452-020-2310-6
[6] Pyragas V, Pyragas K. State-dependent act-and-wait time-delayed feedback control algorithm. Commun Nonlinear Sci Numer Simul 2019;73:338–50. https://doi.org/10.1016/j.cnsns.2019.02.018
[7] Chang A, Bienfang JC, Hall GM, Gardner JR, Gauthier DJ. Stabilizing unstable steady states using extended time-delay autosynchronization. Chaos 1998;8(4):782–90. https://doi.org/10.1063/1.166357
[8] Ahlborn A, Parlitz U. Controlling dynamical systems using multiple delay feedback control. Phys Rev E 2005;72:016206. https://doi.org/10.1103/PhysRevE.72.016206
[9] Hövel P, Dahlem MA, Dahms T, Hiller G, Schöll E. Time-delayed feedback control of delay-coupled neurosystems and lasers. IFAC Proc Vol 2009;2:235–40. https://doi.org/10.3182/20090622-3-UK-3004.00045.
[10] Amster P, Alliera C. Control of pyragas applied to a coupled system with unstable periodic orbits. Bull Math Biol 2018; https://doi.org/10.1007/s11538-018-0492-5

[11] Pastor VE, González GA. Oscillating delayed feedback control schemes for stabilizing equilibrium points. Heliyon 2019;5(6). https://doi.org/10.1016/j.heliyon.2019.e01952
[12] Costa D, Pavlovskaia E, Wiercigroch M. Feedback control of chaos in impact oscillator with multiple time-delays. Chaos Solit Fractals 2024;181:114570. https://doi.org/10.1016/j.chaos.2024.114570
[13] Kemnang Tsafack A, Kengne R, Cheukem A, Mboupda Pone J, Kenne G. Chaos control using self-feedback delay controller and electronic implementation in IFOC of 3-phase induction motor. Chaos: Theory Appl 2020;2:40–8.
[14] Hu Y, Hu G. Stabilization and chaos control of an economic model via a time-delayed feedback scheme. Mathematics 2023;11(13). https://doi.org/10.3390/math11132994
[15] Guan Z, Feng G, Chen C. A stabilization method of chaotic systems based on full delayed feedback controller design. Phys Lett A 2006;348(3):210–21. https://doi.org/10.1016/j.physleta.2005.08.061
[16] Lu J, Ma Z, Li L. Double delayed feedback control for the stabilization of unstable steady states in chaotic systems. Commun Nonlinear Sci Numer Simul 2009;14(7):3037–45. https://doi.org/10.1016/j.cnsns.2008.10.026
[17] Kokame H, Hirata K, Konishi K, Mori T. Difference feedback can stabilize uncertain steady states. IEEE Trans Autom Control 2001;46(12):1908–13. https://doi.org/10.1109/9.975474
[18] Kuznetsov NV, Leonov GA, Shumafov MM. A short survey on pyragas time-delay feedback stabilization and odd number limitation. IFAC Pap 2015;48(11):706–9. https://doi.org/10.1016/j.ifacol.2015.09.271
[19] Shumafov MM. Leonov's method of nonstationary stabilization in the theory of linear control systems. J Phys Conf Ser 2021;1864, 012067. https://doi.org/10.1088/1742-6596/1864/1/012067
[20] Konishi K, Kokame H, Hara N. Delayed feedback control based on the act-and-wait concept. Nonlinear Dyn 2011;63:513–19. https://doi.org/10.1007/s11071-010-9819-y
[21] Leonov GA, Shumafov MM. Pyragas stabilizability of unstable equilibria by nonstationary time-delayed feedback. Autom Remote Control 79 2018;6:1029–39. https://doi.org/10.1134/S0005117918060048
[22] Elaydi S. An Introduction to Difference Equation. Springer-Verlag; 1996.
[23] Boccaletti S, Grebogi C, Lai Y-C, Mancini H, Maza D. The control of chaos: theory and applications. Phys Rep 2000;329(3):103–97. https://doi.org/10.1016/S0370-1573(99)00096-4
[24] Tsuneda A. A GALLERY OF ATTRACTORS FROM SMOOTH Chuas's equation. Int J Bifurc Chaos 2005;15 (1):1–49. https://doi.org/10.1142/S0218127405011990
[25] Galias Z. On rigorous integration of continuous piecewise linear systems. Commun Nonlinear Sci Numer Simul 2022;107:106109. https://doi.org/10.1016/j.cnsns.2021.106109
[26] Matsumoto T, Chua L, Komuro M. The double scroll. IEEE Trans Circuits Syst 1985;32 (8):797–818. https://doi.org/10.1109/TCS.1985.1085791
[27] Belykh VN, Barabash NV, Belykh I. The hidden complexity of a double-scroll attractor: analytic proofs from a piecewise-smooth system. Chaos Interdiscip J Nonlinear Sci 2023;33(4):043119. https://doi.org/10.1063/5.0139064
[28] Bartissol P, Chua LO. The double hook (nonlinear chaotic circuits). IEEE Trans Circuits Syst 1988;35(12):1512–22. https://doi.org/10.1109/31.9914
[29] Galias Z. Chaos in the chua's circuit double-hook attractor. Proc IEEE Int Symp Circuits Syst 2022;:114–17 https://doi.org/10.1109/ISCAS48785.2022.9937905
[30] Leonov GA, Shumafov MM. Stabilization of controllable linear systems. Nonlinear Dyn Syst Theory 2010;10:235–68.
[31] Leonov GA, Shumafov MM, Kuznetsov NV. Delayed feedback stabilization and the Huijberts - Michiels - Nijmeijer problem. Differ Eq 2016;52:1707–31. https://doi.org/10.1134/S0012266116130036
[32] Leonov GA. Pyragas stabilizability via delayed feedback with periodic control gain. Syst Control Lett 2014;69:34–7. https://doi.org/10.1016/j.sysconle.2014.04.001